ULTRON Tech Stack

Tools we use today

Tool	What it is	What we use it for
Next.js 15	React web framework	The ULTRON web app and all its pages
React 19	UI library	Every screen and component
TypeScript	JavaScript with types	All front-end and back-end code
three.js with React Three Fiber	3D graphics library	The 3D digital twin of each machine
SVG machine kit	Our own drawing components	2D machine drawings and the operator HMI view
Hono on Node.js 22	Lightweight API server	Every API route, sensor evaluation and the always-on plant
PostgreSQL on Supabase	Hosted relational database	All plant, machine, sensor, reading and maintenance data
MQTT.js	MQTT client library	Receiving sensor readings from gateways and sending commands back
EMQX Public Cluster	Public MQTT broker	Local testing without plant hardware
Server-Sent Events	Browser push over HTTP	Streaming live readings to open pages
Vercel	Web hosting platform	Hosting the web app
GitHub	Code hosting	Source control for the ORCAS repository
npm, esbuild, tsx	Build and run tools	Installing packages, bundling and running the API
ESLint	Code checker	Catching code mistakes before they ship

Future tools

Tool	What it is	What we will use it for
Amazon RDS for PostgreSQL	Managed PostgreSQL on AWS	Plant setup, sensor definitions, users, work orders and alerts in production
Private EMQX	Production MQTT broker	Receiving data from every plant gateway
Apache Kafka or Redpanda (Amazon MSK or Kinesis)	Stream buffer	Holding incoming sensor data and feeding it to the databases in batches
Redis or Valkey on Amazon ElastiCache	In-memory database	The latest live value of every sensor, for instant screen updates
ClickHouse (or QuestDB)	Time-series database	Every 1-second sensor value, plus 1-minute, 1-hour and 1-day summaries for trend charts
Amazon S3 with Glacier Instant Retrieval	Cloud file storage	Vibration waveform recordings, years of old sensor data, inspection photos and PDF manuals

Tool	What it is	What we will use it for
Apache Parquet with ZSTD	Compressed file format	Storing waveform arrays and old sensor data in a fraction of the space
Amazon ECS Fargate or App Runner	Managed container hosting	Running the web app, API and stream worker without managing servers
Application Load Balancer and ACM	Load balancer and SSL certificates	Secure HTTPS access to ULTRON
Isolation Forest (scikit-learn)	Anomaly detection algorithm	First ML model: flagging unusual sensor patterns
LightGBM	Tree-based ML models	Predicting expected values and classifying faults
SHAP	Model explanation library	Showing which sensors drove each ML result
LSTM sequence models	Deep learning for time series	Forecasting sensor values, at a later stage
AI orchestration pipeline	Workflow that routes ULTRON's AI tasks to frontier models	Powering ULTRON's AI features until our own ML and AI models are built and trained

Until ULTRON's own ML and AI models are built, trained and validated, the orchestration pipeline uses frontier models such as Claude and DeepSeek. The planned ML models above take over as each one is ready.

For testing, each layer has a free option: Supabase or Neon for PostgreSQL, Redis Cloud, Aiven for ClickHouse, Cloudflare R2 or MinIO for S3, and EMQX Serverless or HiveMQ for MQTT.

AI tools we use

Tool	What it is	What we use it for
Claude Code	Anthropic's AI coding agent, in the Claude desktop app	Writing, reviewing and testing ULTRON code
Claude Opus 5.5	Anthropic AI model	Main coding model; also reads onboarding PDFs and draws machines from photos
Claude Haiku	Small, low-cost Anthropic model	Testing cheap PDF extraction
Claude project skills	Reusable instructions for Claude	PDF extraction, 2D and 3D machine drawings, plant onboarding
graphify	Codebase knowledge graph	Helping AI agents find the right code quickly
OpenAI Chatgpt	AI coding agent	Research and Documentation
Figma and Canva connectors	Design tools linked to Claude	Design work and visuals
Claude Docs	Shared documents made by Claude	Specs and reports such as this one
Claude API	Anthropic's model API	Powering plant onboarding: reading uploaded PDFs and photos and turning them into plant data and drawings
OpenCode	Open-source AI coding agent	Coding with a choice of models alongside Claude Code
DeepSeek API	AI models from DeepSeek	AI Model to power chatbot and other onboarding processes